Complete Walkthrough: Hamiltonian Learning Mini-Project

Every concept, every step, every design choice — explained in full

Prepared for: meeting with Prof. Sudipto Singha Roy

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This document walks through everything behind the mini-project report, from the original assignment to the final numbers, explaining every concept used along the way so you can answer any question Prof. Roy asks — including ones about things that didn't make it into the formal report. It is organized as: (1) the big picture, (2) the literature background, (3) every physics and math concept used in the implementation, (4) a line-by-line walkthrough of what the code actually does, (5) how to read each result, (6) honest weak points, and (7) a rehearsal Q&A section.

1. The Big Picture

1.1 What Prof. Roy actually asked for

His email posed two linked questions: (a) can we design and implement a desired Hamiltonian on quantum hardware, and (b) if the realized Hamiltonian differs from the intended one due to hardware errors, can we detect and characterize that difference? He asked for a literature survey on two topics — Hamiltonian learning and randomized benchmarking — and gave one paper (Van Kirk, Cotler, Huang & Lukin, arXiv:2212.06084) and a YouTube video as a starting point, with three weeks for a presentation.

1.2 Why a mini-project, and why now

With the deadline close and no concrete deliverable yet built, the decision was made to convert the literature survey into a working numerical demonstration: take the exact measurement framework from the paper Prof. Roy supplied, and use it to directly answer his question — can restricted, realistic randomized measurements detect that a device's Hamiltonian has drifted from its design? This produces something concrete to show (code, validated results, plots) rather than only a reading summary.

1.3 One-sentence summary of what was built

A small 4-qubit spin chain was simulated twice — once as intended, once with injected calibration errors — and a measurement scheme that uses only a single randomly-rotated, globally-shared control knob (no individual qubit addressing) was used to try to detect, purely from measurement statistics, that the second Hamiltonian is not the first.

2. Background Concepts: Why This Problem Exists

2.1 Hamiltonian learning

A Hamiltonian H is the operator that generates a quantum system's time evolution: it determines the energy levels and the dynamics (via the Schrodinger equation). When a Hamiltonian is 'simulated' on quantum hardware — for instance a chain of qubits engineered to behave like a chain of interacting spins — the device is supposed to realize a specific target H. But real hardware has calibration error, cross-talk between neighboring qubits, drift over time, and other imperfections. The Hamiltonian the device actually realizes, call it H', is therefore not exactly H. Hamiltonian learning is the general name for techniques that use measurement data taken from the device to estimate what H' actually is, rather than assuming H' = H because that's what was intended.

This matters because as quantum simulators get larger, we lose the ability to classically simulate them to check their answers — so we need measurement-based ways to verify the device is doing physics we understand and intended, not something else.

2.2 Randomized benchmarking (RB)

Randomized benchmarking is a family of protocols, originally developed to measure gate/circuit error rates, that apply randomized sequences of operations and look at how a signal (e.g. fidelity) decays with sequence length. Its defining virtue is that it is scalable (doesn't need exponentially many measurements) and robust to state-preparation-and-measurement (SPAM) errors — i.e. it doesn't require perfectly trusted state preparation or readout, because the randomization 'averages away' many systematic errors. Hamiltonian learning borrows this randomization philosophy: rather than needing full tomography (measuring literally everything), use randomized measurement strategies to extract just the information needed (e.g. specific Hamiltonian coefficients) efficiently and in a way that is robust to SPAM-type errors.

2.3 Why these two topics are paired

Randomized benchmarking is not itself a Hamiltonian-reconstruction technique — it is a measurement philosophy. Hamiltonian learning is the end goal (know the Hamiltonian). The randomized-measurement / classical-shadow toolbox (which includes RB-style ideas) is the means: a scalable, robust way of extracting the needed information. This project sits exactly at that junction — it uses an RB-style randomized measurement protocol as the data-collection method for a Hamiltonian-verification task.

3. The Four Key Papers — What Each One Actually Says

3.1 Wiebe, Granade, Ferrie & Cory — Hamiltonian Learning and Certification Using Quantum Resources (PRL 2014)

Problem: as quantum simulators get too large to check against a classical computer, how do you certify that a simulator is actually running the Hamiltonian you intended? Their answer: use a second, small, trusted quantum device together with Bayesian inference (an iterative 'update your belief as data arrives' statistical procedure) to efficiently pin down the large untrusted simulator's Hamiltonian. They demonstrated this numerically on large frustrated Ising models. This paper essentially founded the modern framing of Hamiltonian learning as a certification tool, which is precisely Prof. Roy's framing.

3.2 Bairey, Arad & Lindner — Learning a Local Hamiltonian from Local Measurements (PRL 2019)

Problem: can you reconstruct a Hamiltonian using only local measurements (the kind actually available on near-term hardware), instead of needing full state tomography (which is exponentially expensive)? Their key insight: if you measure a state that is an eigenstate of H (e.g. the ground state), then for any local operator O, the expectation value of the commutator [H, O] must be exactly zero in that eigenstate (because eigenstates don't change in time — nothing evolves, so every observable's rate of change is zero). This gives you a large set of linear equations relating the unknown Hamiltonian coefficients to measurable local quantities, which you can solve with ordinary linear algebra. Because only local measurements are needed, the computational and measurement cost scales only linearly with system size, instead of exponentially. This is one of the most practical and widely cited Hamiltonian-learning methods.

3.3 Huang, Tong, Fang & Su — Learning Many-Body Hamiltonians with Heisenberg-Limited Scaling (PRL 2023)

Problem: what is the absolute best possible sample/time efficiency any Hamiltonian-learning algorithm could achieve? Standard statistical estimation (averaging over many random measurements) typically needs resources scaling like 1/epsilon^2 to reach precision epsilon — this is called the 'standard quantum limit'. This paper proves you can do better: O(1/epsilon) scaling — the 'Heisenberg limit' — by using the Hamiltonian's own time evolution as a resource (letting the system evolve for carefully chosen times, rather than just repeatedly re-measuring). They also prove a matching lower bound, meaning no algorithm can do fundamentally better than this. This is the theoretical ceiling for the field.

3.4 Van Kirk, Cotler, Huang & Lukin — Hardware-Efficient Learning of Quantum Many-Body States (arXiv:2212.06084, 2022) — the paper Prof. Roy sent

Important distinction to be ready to state clearly: this paper is NOT about Hamiltonian learning. It is about state-property estimation — given an unknown quantum state (not a Hamiltonian), estimate many of its properties (expectation values of many observables) efficiently. This generalizes a technique called 'classical shadow tomography'. The reason it is still directly useful for Prof. Roy's question is explained fully in Section 4 below: its measurement machinery is exactly what this mini-project repurposed to attack the Hamiltonian-verification problem.

4. Deep Dive: The Van Kirk et al. Paper and Classical Shadows

4.1 Classical shadow tomography — the starting idea (Huang, Kueng & Preskill, 2020)

Ordinary quantum state tomography — fully reconstructing an unknown state rho — requires a number of measurements that grows exponentially with the number of qubits, because rho has exponentially many independent parameters. Classical shadows sidestep this when you don't need the full state, only expectation values of a limited (but possibly large) number of observables. The idea: repeatedly (1) apply a randomly chosen unitary U to the state, (2) measure in the computational basis to get an outcome b, and (3) classically compute a 'snapshot' operator from (U, b) using a fixed formula. Averaging many such snapshots gives an unbiased estimator of rho itself (and hence of tr(rho O) for any observable O), using only O(log M / epsilon^2) measurements to estimate M different observables to precision epsilon each — a logarithmic, not linear or exponential, dependence on the number of properties you want to know.

The classic version of this protocol (Huang-Kueng-Preskill) applies an INDEPENDENT random rotation to every qubit. The snapshot formula for that specific ensemble has a simple closed form: for each qubit, the snapshot is 3*U_dagger|b><b|U - I (a simple linear rescaling of the measured projector), and the full-system snapshot is just the tensor product of the single-qubit snapshots. This closed-form baseline is implemented in the project's code as the 'full control' comparison method.

4.2 The restriction Van Kirk et al. address: you may not have per-qubit control

Many real devices — Rydberg-atom arrays, certain trapped-ion and cavity-QED setups — can only apply the SAME control operation to every particle simultaneously (e.g. one shared laser pulse hits the whole array identically). You cannot independently rotate qubit 3 differently from qubit 7. The Huang-Kueng-Preskill closed-form snapshot formula assumes independent per-qubit randomization and does not directly apply here. Van Kirk et al. generalize the classical-shadow idea to work for ANY restricted set of 'implementable unitaries' — including this 'one shared global rotation' case — not just independent per-qubit rotation.

4.3 The visible space / invisible space formalism

Given whatever restricted set of unitaries V you can actually apply (here: identical rotation on every qubit) plus computational-basis measurement, define the 'visible space' as the span of every operator that your measurement statistics can, in principle, distinguish — formally, the span of all operators V_dagger|b><b|V over every allowed V and outcome b. Any observable O living inside this span can have tr(rho O) estimated from your data, no matter how restricted V is. Any observable orthogonal to this span (the 'invisible space') can NEVER be learned from this measurement scheme, regardless of how many shots you take — it's a structural, not a statistical, limitation.

Their key quantitative finding: even for the most restricted case they study (identical rotation on every qubit — exactly what this project uses), the visible space is still exponentially large in the number of qubits — drastically restricted control still leaves you able to learn an enormous amount.

4.4 The general estimator ('kernel') formula

For a target observable O inside the visible space, you can write it as a linear combination of the V_dagger|b><b|V operators. The coefficients of that combination, viewed as a function of (V,b), are called the kernel K^O(V,b). Then tr(rho O) is simply the average of K^O(V,b) over your collected (V,b) measurement data — this generalizes the Huang-Kueng-Preskill snapshot-and-average procedure to arbitrary restricted ensembles. Their Theorem 1 shows you keep the same favorable O(log M / epsilon^2) scaling in the number of measurements needed for M target observables, even under this generalization.

4.5 Two efficiency tricks from the paper (not used numerically here, but worth knowing)

- Bias-variance tradeoff: instead of estimating your true target O exactly, estimate a nearby, 'easier' operator O-tilde that has lower statistical variance, as long as the resulting bias stays inside your error budget.
- Adaptive sampling: instead of sampling the allowed unitaries uniformly at random, bias the sampling distribution toward whichever unitaries are most informative for the specific observables you care about, cutting the number of shots needed.

4.6 Their two demonstrated applications

- Energy density of a 2+1D U(1) lattice gauge theory: they show the Hamiltonian's own local terms (plaquette and link operators) happen to live inside the visible space of global-SU(2) control, so even this crude control is enough to extract local energy densities — a direct precedent for what this mini-project does with a spin chain's own coupling terms.
- Classifying topological order (toric code vs. trivial phase) in a 200-qubit system using only global SU(2) measurements fed into an unsupervised machine-learning step (kernel PCA) — demonstrating these restricted measurements carry enough information for genuinely hard many-body classification tasks, not just simple observables.

4.7 The crucial reframe this project makes

Van Kirk et al.'s paper answers: 'given a state, what can restricted measurements tell us about it?' Prof. Roy's question is: 'given a Hamiltonian realized on hardware, can restricted measurements tell us whether it matches the design?' This project connects the two by noting that if you measure the ground state of whatever Hamiltonian the device actually realizes, and you use the paper's restricted-measurement kernel estimator to read off the expectation values of the intended Hamiltonian's own local terms, you get exactly the information needed to compare 'what the device is doing energetically' against 'what it was designed to do' — turning a state-property-estimation tool into a Hamiltonian-verification tool.

5. Every Concept Used in the Implementation, Explained

5.1 The model Hamiltonian: an XXZ / Heisenberg spin chain

A chain of n 'spins' (here realized as qubits) is modeled with nearest-neighbor coupling terms. The Pauli matrices X, Y, Z are the standard 2x2 operators representing spin components along three orthogonal axes. The Hamiltonian used is:

$$H0 = \sum_i J\,(X_i X_{i+1} + Y_i Y_{i+1} + \Delta\, Z_i Z_{i+1}) + h \sum_i Z_i$$

With Delta = 1 this is the isotropic Heisenberg model — fully rotationally symmetric under SU(2) (simultaneous rotation of all spins about any axis leaves the physics unchanged). This symmetry turns out to matter a great deal for the results (see Section 6.4). The small extra 'h' term is a weak uniform field along Z, included so the Hamiltonian isn't perfectly degenerate and has a well-defined, non-trivial ground state.

Why 4 qubits specifically

A system of n qubits has a Hilbert space of dimension 2^n. At n=4, that's 16 — small enough that the Hamiltonian can be written out as an explicit 16x16 matrix and exactly diagonalized on a laptop in milliseconds, with no approximation. This was essential for finishing the whole pipeline (model, measurement simulation, estimator construction, validation, and the full experiment) in a single day.

5.2 Pauli strings and the operator basis

For n qubits, a 'Pauli string' is a tensor product of one single-qubit Pauli operator (I, X, Y, or Z) per site — e.g. for n=4, the string 'XIZI' means X on qubit 0, identity on qubit 1, Z on qubit 2, identity on qubit 3. There are exactly 4^n such strings (4 choices per site, n sites) — for n=4 that's 256. These 256 operators form a complete orthogonal basis for the space of all possible 4-qubit Hermitian operators (which also has dimension 4^n = 256, since a 16x16 Hermitian matrix has exactly 256 independent real parameters). Every operator used in this project — the

Hamiltonian terms, the measurement-response map, the dual/kernel operators — is expressed as a linear combination of these 256 basis operators. This is the natural 'coordinate system' for working with multi-qubit operators.

5.3 Ground state and exact diagonalization

The ground state of a Hamiltonian is its lowest-energy eigenstate — physically, the state the system settles into at zero temperature, and the state many quantum simulation protocols are specifically designed to prepare (e.g. via adiabatic evolution). 'Exact diagonalization' means literally computing all eigenvalues and eigenvectors of the Hamiltonian matrix using standard linear algebra (here, scipy's eigh routine, which is appropriate for Hermitian matrices). This is only feasible because the Hilbert space is small (dimension 16); for larger systems this becomes exponentially expensive, which is exactly why practical Hamiltonian-learning methods avoid needing full diagonalization or full tomography of large systems.

5.4 The noise model: what 'realized Hamiltonian' means here

To simulate a miscalibrated device, each bond coupling J is multiplied by (1 + epsilon * delta_i), where delta_i is a fixed random number (drawn once, between -1 and 1, different per bond) and epsilon is a single knob controlling how large the miscalibration is overall. epsilon = 0 reproduces the intended Hamiltonian exactly; larger epsilon means a more badly miscalibrated device. The state actually probed in the experiment is the exact ground state of this noisy Hamiltonian — representing what a real device would produce if it tried to prepare the intended ground state but its couplings were actually off by this pattern.

5.5 The restricted-control measurement model

Haar-random SU(2) rotations

SU(2) is the group of all 2x2 unitary matrices with determinant 1 — physically, every possible way to rotate a single qubit's state on the Bloch sphere. 'Haar-random' means sampled uniformly over this group, with no direction preferred over any other — the proper notion of 'a completely random rotation' for a continuous group, analogous to picking a uniformly random point on a sphere. In the code, such a matrix is generated via a QR decomposition of a random complex (Ginibre) matrix, a standard, numerically robust way to sample Haar-random unitaries.

Applying the SAME rotation to every qubit

The restricted-control constraint is implemented by building one random single-qubit unitary u, then forming the full n-qubit operation as U = u tensor u tensor u tensor u — i.e. literally the same rotation repeated on every qubit, never independent per-qubit rotations. This directly encodes the 'global control only' physical restriction central to the Van Kirk et al. framework and to real devices like Rydberg arrays.

Computational-basis measurement and the Born rule

After applying U, the (simulated) qubits are measured in the standard computational basis (i.e. each qubit is measured as '0' or '1'). The probability of observing a particular outcome string b is given by the Born rule: probability = |<b| U |psi>|^2 for a pure state psi, or the corresponding diagonal entry of U*rho*U_dagger for a mixed state. The code samples an outcome stochastically according to exactly these probabilities, which is what makes this a faithful simulation of a real randomized measurement rather than just a calculation of expectation values.

5.6 Building the estimator: the measurement-response map M and its inverse

Why you need a 'response map' at all

A single measurement shot (one random U, one measured outcome b) tells you almost nothing on its own — it's just one bit of information. Classical shadows work by defining a fixed mathematical transformation that, when applied to the raw (U,b) outcome and then AVERAGED over many shots, reconstructs an unbiased estimate of whatever you want (the state, or specific expectation values). The object that needs to be inverted to find this transformation is the superoperator M, defined as M(O) = (average over random U of) [U_dagger * (dephase the diagonal of U O U_dagger) * U], where 'dephase' means zero out every off-diagonal matrix entry, mimicking what a projective measurement does to a density matrix. M is a fixed, state-independent property of the measurement scheme alone — once you know M (and its inverse), you can estimate tr(rho O) for ANY state rho, using the same formula.

Why Monte Carlo rather than a closed formula

Van Kirk et al. derive M's inverse analytically for their restricted ensemble using representation theory (specifically Schur-Weyl duality, a deep but intricate mathematical tool for decomposing how tensor-power group actions behave). Deriving that closed form was not feasible within the one-day timeline. Instead, M was estimated numerically: thousands of random SU(2) matrices were sampled, and for each one, the action of M on every one of the 256 Pauli-basis operators was computed and averaged. This produces a 256x256 numerical matrix representation of M (in the Pauli basis) that converges to the true M as the number of Monte Carlo samples grows — this is a legitimate, if less elegant, substitute for the exact representation-theoretic derivation, and it was explicitly validated (Section 5.8) before being trusted.

Why a pseudo-inverse, not an ordinary inverse

Because the control is restricted, M is NOT invertible in the ordinary sense — it has a non-trivial null space, exactly corresponding to the 'invisible space' from Section 4.3 (operators that this measurement scheme simply cannot see, numerically confirmed to have rank 104 out of 256 possible directions). The Moore-Penrose pseudo-inverse (computed via `numpy.linalg.pinv`) is the standard way to 'invert' such a rank-deficient matrix: for any target vector, it returns the minimum-norm solution among all solutions that are consistent with the visible part of the map, automatically producing zero component along directions the measurement can't see. This is exactly the numerical analogue of 'projecting onto the visible space' that the paper performs analytically.

Dual operators and the final estimator formula

For each Hamiltonian term O you want to estimate (e.g. X_0 X_1), the 'dual operator' X_dual = M^{-1}(O) is computed once (via the pseudo-inverse solve). Then, for a single measurement shot (U, b), the quantity <b| U X_dual U_dagger |b> is computed; averaging this quantity over many independent shots gives an unbiased estimate of tr(rho O) — this is precisely the kernel estimator K^O(U,b) from Section 4.4, built numerically instead of analytically.

5.7 The full-control baseline (for comparison)

To know whether the restricted-control scheme is actually costing anything, the project also implements the standard, well-established Huang-Kueng-Preskill classical-shadow protocol: an INDEPENDENT Haar-random rotation on every qubit (not the same one shared across all qubits), with its simple closed-form per-qubit snapshot 3*U_dagger|b><b|U - I, tensored across qubits. This needs no Monte Carlo frame construction at all, since its inverse channel is known analytically.

5.8 Validation: why it was done, and what it proved

Before trusting any of the above machinery on the real experiment, it was tested on a case where the right answer is already known exactly: the clean (epsilon=0) ground state. Using 20,000 simulated restricted-control measurement shots, every one of the 13 tracked Hamiltonian-term expectation values was estimated and compared against its exact value (computed directly from the ground-state wavefunction via ordinary matrix algebra, with no randomness involved). All 13 estimates agreed with the exact values within about 1-2 standard errors — the statistically expected level of agreement for an unbiased estimator — confirming that the Monte Carlo frame construction, the pseudo-inverse dual-operator calculation, and the final kernel-averaging formula were all implemented correctly before they were used for anything that actually mattered.

5.9 The experiment design: two sweeps

Sweep 1 — detectability vs. noise strength epsilon

For several values of epsilon (0 up to 0.5), the ground state of the correspondingly noisy Hamiltonian was prepared, and its 13-term Hamiltonian profile was estimated using a fixed budget of 8000 restricted-control measurement shots, repeated 6 independent times to get a mean and a standard deviation (not just a single noisy sample). The key quantity tracked is the Euclidean distance (L2 norm) between the 13-dimensional vector of estimated term values and the 13-dimensional vector of values the INTENDED (epsilon=0) Hamiltonian's own ground state would give — i.e. 'how far does the measured energy profile look from what we designed for.' This distance was compared against

the true (noise-free, exactly computed) distance, and against the expected statistical noise floor set by the finite shot budget.

Sweep 2 — estimation accuracy vs. shot count, at fixed noise

At a fixed, moderately noisy Hamiltonian (epsilon=0.2), the root-mean-square error (RMSE) of the estimated 13-term profile (against the TRUE profile of that noisy state, not the ideal one) was tracked as the number of shots increased from 200 to 10,000, for both the restricted-control estimator and the full-control baseline (capped at 3000 shots for the baseline purely to keep runtime manageable, since its per-shot cost is higher).

Why Euclidean distance / RMSE as the metric

Both are standard ways to collapse a vector of 13 separate small errors into one number representing 'how far off, overall.' RMSE additionally normalizes by the number of terms, making it comparable in meaning even if the term count had differed.

6. How to Read Every Result

6.1 Visible space: 104 out of 256

This is a purely structural fact about the measurement scheme, independent of any noise or shot count: under global-rotation-only control, exactly 104 of the 256 possible 4-qubit Pauli-operator directions are, in principle, ever learnable — the rest are permanently invisible to this measurement scheme, no matter how much data is collected. This matches the Van Kirk et al. paper's qualitative claim that restricted control still leaves access to a large (though not complete) fraction of operator space.

6.2 The detection-vs-epsilon plot

At epsilon=0 (no noise), the estimated distance from the ideal profile is NOT exactly zero — it sits around 0.09, matching almost exactly the expected statistical noise floor from finite shots (shown as the dotted gray line). This is exactly what should happen: even a perfectly calibrated device will show nonzero apparent deviation purely from measurement (shot) noise. As epsilon grows, the TRUE deviation (red dashed line, computed exactly, no measurement noise) grows steadily from zero. The ESTIMATED deviation (blue line) tracks the true deviation reasonably well once the true deviation exceeds the noise floor — around epsilon of about 0.2 and above — meaning that at this shot budget, miscalibration below roughly epsilon~0.1-0.2 would be statistically indistinguishable from a perfectly calibrated device, while miscalibration above that threshold is clearly detectable. This defines a concrete 'detection threshold' for this system size and shot budget — exactly the kind of number a real calibration-verification protocol would need to report.

6.3 The RMSE-vs-shots plot

Both the restricted and full-control schemes show RMSE decreasing as shot count increases, consistent with the general expectation that measurement noise shrinks like 1/sqrt(N) for N shots (shown as the dotted reference line). The curves are not perfectly smooth at low shot counts because each point here comes from a single trial rather than an average over many trials (a limitation, discussed below) — but the overall downward trend, and the crossing below the full-control baseline at larger N, are the real, meaningful signal.

6.4 Why the restricted scheme matches or beats the full-control baseline

This is the most surprising and most important result to be ready to explain. Intuitively, restricting control should only ever make things harder, never easier. The resolution: the specific Hamiltonian terms being estimated here (the XX+YY+ZZ exchange couplings) are themselves symmetric under simultaneous rotation of all spins (an SU(2)-symmetric, or 'Heisenberg-type,' quantity). The restricted measurement scheme used here applies exactly this same kind of simultaneous, identical rotation to every qubit — i.e. the measurement ensemble's own symmetry matches the symmetry of the quantities being estimated. When a measurement ensemble's symmetry is well matched to the observable's symmetry, statistical noise can genuinely be LOWER than for a generic, unstructured (independent-per-qubit) randomization — this is not a numerical artifact, but a real statistical effect, and it is also exactly what Van

Kirk et al. observe in their own lattice-gauge-theory example. It is an important caveat that this favorable result is specific to symmetric Hamiltonians and may not hold for Hamiltonians without this structure (see Section 7).

7. Honest Weak Points — Know These Before He Asks

- System size (n=4) is small purely for same-day tractability; nothing here demonstrates how the visible-space fraction or the detection threshold scale as the system grows — that is an open question, not yet addressed.
- The measurement-response map M was built via Monte Carlo sampling (6000 random rotations), not the exact closed-form representation-theory (Schur-Weyl) derivation the paper itself uses for symmetric cases — this introduces a small, controlled amount of extra numerical noise into the dual operators themselves, on top of ordinary measurement shot noise. It was validated against ground truth to confirm this noise is small enough not to bias results, but it was not separately quantified against shot noise.
- Only one random noise pattern (the per-bond delta_i values) was used, scaled by epsilon; a different random pattern could in principle show a different detection threshold. No averaging over multiple independent noise realizations was done.
- The 'device state' is the EXACT ground state of the noisy Hamiltonian, obtained by direct diagonalization — a convenient stand-in, not a simulation of an actual physical state-preparation protocol (e.g. adiabatic sweep or variational circuit) under realistic noise, which would behave somewhat differently.
- The favorable sample-efficiency result (restricted beats/matches full control) is specific to this SU(2)-symmetric Heisenberg Hamiltonian; it is an open question whether it holds for Hamiltonians that break this symmetry (e.g. anisotropic or non-Hermitian Hamiltonians).
- Sweep 2's RMSE values come from single trials rather than averages over repeated trials, so the curve has visible statistical jitter at low shot counts, though the overall downward trend is real.

8. Code Walkthrough

8.1 core.py

Defines the Pauli matrices, a helper to build a Pauli string operator on n qubits via tensor (Kronecker) products, a function to enumerate all 4^n Pauli strings, the XXZ Hamiltonian builder, a function returning every individual local term of that Hamiltonian as its own operator (what is actually being estimated), and ground-state / thermal-state helpers using scipy's exact Hermitian eigensolver.

8.2 shadows.py

Contains: a Haar-random SU(2) sampler; the global-rotation operator builder (U = u tensor u ... tensor u); a function to simulate one restricted-control measurement shot (sample a random rotation, apply it, sample a measurement outcome via the Born rule); the RestrictedFrame class, which on construction runs the Monte Carlo estimate of the measurement-response map M in the Pauli basis, and exposes a dual_operator() method (the pseudo-inverse solve) and a kernel_estimate() method (average the kernel over a batch of shots); and finally the full-control baseline functions implementing the standard independent-per-qubit classical-shadow protocol for comparison.

8.3 validate.py

Builds the frame, computes exact ground-truth expectation values for the clean Hamiltonian's ground state, draws 20,000 restricted-control shots, and prints a side-by-side comparison table of exact vs. estimated values with their statistical uncertainties — the sanity check described in Section 5.8.

8.4 experiment.py

The main script: builds the intended Hamiltonian and its term list, fixes the random noise pattern, builds the restricted-control frame once (reused for everything downstream), precomputes the 13 dual operators once, then runs Sweep 1 (noise-detection, 6 repeats per epsilon value) and Sweep 2 (shot-count scaling, both schemes), printing results and saving everything to results.json for the plotting step.

8.5 plot.py

Reads results.json and produces the three figures used in both the report and this document: the detection-vs-epsilon plot, the RMSE-vs-shots plot, and the visible/invisible space bar chart.

9. Rehearsal: Likely Questions and Suggested Answers

Q: Isn't the Van Kirk et al. paper about state learning, not Hamiltonian learning — why did you use it for this?

A: Correct, it's a state-property-estimation paper, not a Hamiltonian-reconstruction paper. I used it because its restricted-control randomized-measurement machinery — the visible-space formalism and the kernel estimator — is measurement-scheme machinery, independent of what you ultimately want to estimate. I repurposed it: instead of estimating generic state properties, I used it to estimate the specific local terms of my own intended Hamiltonian, which turns a state-learning tool into a Hamiltonian-verification tool.

Q: Why global rotation specifically, instead of independent per-qubit rotation?

A: Because that's the physically realistic constraint on some real hardware platforms — Rydberg arrays and similar setups often only allow one shared control field across the whole array, not individual qubit addressing. It's also literally the restricted-control case Van Kirk et al. analyze.

Q: Why didn't you derive the closed-form kernel the paper uses?

A: Their closed form relies on representation theory (Schur-Weyl duality) specific to their symmetric observables, which would have taken considerably longer to derive correctly than the one day available. I instead constructed the measurement-response map numerically via Monte Carlo and validated it against exact ground truth, which is a legitimate substitute — slower and less elegant, but verifiably correct for this system size.

Q: Why only 4 qubits?

A: Purely to keep the whole pipeline — exact diagonalization, frame construction, estimator validation, and the full experiment — tractable within a single day, since the Hilbert space (dimension 16) and operator space (dimension 256) are small enough for exact, fast numerics. Scaling to larger systems is listed explicitly as a next step.

Q: How do you know your noise-detection result isn't just numerical noise in your estimator?

A: Two reasons: first, the estimator was validated against exact ground truth on the noise-free case before being trusted for anything else. Second, in the main experiment, the estimated detection threshold matches closely with the independently, exactly computed noise floor expected from the shot budget alone (no randomness in that calculation) — the two numbers agreeing is strong evidence the estimator is behaving correctly rather than producing an artifact.

Q: Why does the restricted scheme do as well as, or better than, full per-qubit control? Shouldn't more control always help?

A: It would for a generic, unstructured problem. But the specific quantities being estimated (Heisenberg exchange couplings) are themselves symmetric under simultaneous rotation of all spins, and the restricted measurement scheme applies exactly that kind of simultaneous rotation. When the measurement ensemble's symmetry matches the observable's symmetry, statistical efficiency can genuinely improve — this is also exactly what Van Kirk et al. find in their own gauge-theory example. It's a real, not spurious, effect, though it may not hold for Hamiltonians without this symmetry.

Q: What would you need to do to make this rigorous enough for a real preprint or paper?

A: Scale to larger n with sparse/Krylov diagonalization; derive the exact closed-form kernel instead of the Monte Carlo approximation; average over many independent noise realizations instead of one fixed pattern; replace the exact-ground-state probe with a simulated, realistically noisy state-preparation protocol; and test Hamiltonians without the Heisenberg symmetry to see if the sample-efficiency advantage persists.

Q: How does this connect to your own research interests / the directions you proposed earlier?

A: Directly: the next proposed step is to apply exactly this restricted-measurement detection protocol to non-Hermitian, PT-symmetric Hamiltonians near an exceptional point — testing whether this kind of measurement

scheme can detect when noise pushes a device across an exceptional point, which connects straight into Prof. Roy's own non-Hermitian physics and eigenstate-thermalization research.

Q: What is a 'dual operator' / why do you need M^{-1}(O) rather than just using O directly?

A: A single measurement shot's raw outcome doesn't equal the observable you want — you need a specific linear transformation of the (random unitary, outcome) pair whose AVERAGE over many shots reconstructs tr(rho O). That transformation is defined through the inverse of the measurement's own response function M. Using O directly, without this correction, would give a biased, physically wrong answer.

Q: Why a pseudo-inverse instead of a normal matrix inverse?

A: Because M is not invertible in the usual sense here — it has a null space exactly equal to the 'invisible space,' the set of operators this restricted measurement scheme cannot see at all. The pseudo-inverse gives the minimum-norm solution consistent with what IS visible, which is the correct, standard way to handle this kind of structurally rank-deficient problem.

10. One-Page Cheat Sheet (for a quick last look before the meeting)

- Hamiltonian learning = inferring the Hamiltonian a device actually realizes from measurement data, instead of trusting the design.
- Randomized benchmarking = the SPAM-robust, scalable randomized-measurement philosophy that Hamiltonian learning borrows its measurement strategies from.
- Van Kirk et al. paper = state-property estimation under restricted control (global rotation only), generalizing classical shadow tomography; its measurement machinery, not its topic, is what this project reuses.
- Visible/invisible space = the structural limit on what any given restricted measurement scheme can ever learn, independent of shot count; here, 104/256.
- Classical shadow = randomize, measure, apply a fixed correction formula, average many shots -> unbiased estimate of any desired expectation value.
- M and its pseudo-inverse = the measurement scheme's own response function and its structurally-aware inverse, needed to turn raw (rotation, outcome) data into an unbiased estimator.
- Main result = restricted-control randomized measurements CAN detect Hamiltonian miscalibration, once the true deviation exceeds the shot-noise floor; and for this symmetric Hamiltonian, restricted control is at least as sample-efficient as full per-qubit control.
- Biggest caveats = small system size, Monte Carlo (not closed-form) kernel, single noise realization, exact-ground-state stand-in for a real noisy state-prep protocol, and a result that may be specific to symmetric Hamiltonians.